

Estimating Nonlinear Insulin-Glucose Dynamics in Critically-ill Patients

Markus Mulvihill

KTH Royal Institute of Technology

August 7th, 2026



2026-08-11

Thesis Presentation

Estimating Nonlinear Insulin-Glucose Dynamics in Critically-ill Patients

Markus Mulvihill

KTH Royal Institute of Technology

August 7th, 2026



1. Supervisor - Martin Lindström
2. Examiner - Ragnar Thobaben
3. Opponent - Divyayan Day

- 1 Motivation
 - The Fundamental Problem Addressed
 - Previous Work
- 2 Results and Contributions
 - Main Results
 - Key Ideas Behind Implementation
- 3 Summary
 - Takeaways
 - Further Reading

2026-08-11

Thesis Presentation

- └ Motivation
 - └ The Fundamental Problem Addressed
 - └ Outline

Outline

- Motivation
 - The Fundamental Problem Addressed
 - Previous Work
- Results and Contributions
 - Main Results
 - Key Ideas Behind Implementation
- Summary
 - Takeaways
 - Further Reading

Fundamentals of Glucoregulation

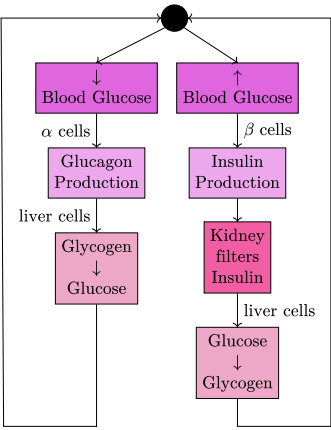


Figure: Flowchart of glucoregulation in healthy individuals.

2026-08-11

- Thesis Presentation
 - Motivation
 - The Fundamental Problem Addressed
 - Fundamentals of Glucoregulation

Fundamentals of Glucoregulation

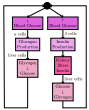
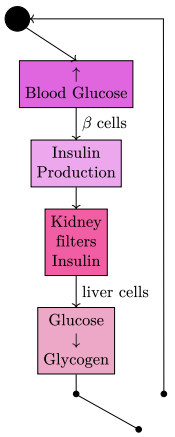


Figure: Flowchart of glucoregulation in healthy individuals.

Insulin Resistance in ICU Patients



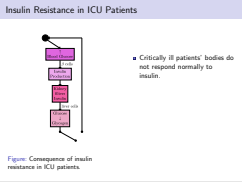
- Critically ill patients' bodies do not respond normally to insulin.

Figure: Consequence of insulin resistance in ICU patients.

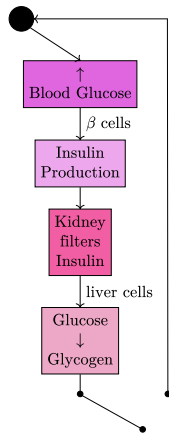
2026-08-11

Thesis Presentation

- └ Motivation
 - └ The Fundamental Problem Addressed
 - └ Insulin Resistance in ICU Patients



Insulin Resistance in ICU Patients



- Critically ill patients' bodies do not respond normally to insulin.
- Stress induced from critical illness increase insulin resistance.

Figure: Consequence of insulin resistance in ICU patients.

2026-08-11

Thesis Presentation

Motivation

The Fundamental Problem Addressed

Insulin Resistance in ICU Patients

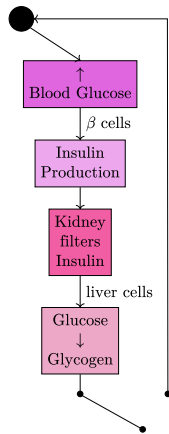
Insulin Resistance in ICU Patients



- Critically ill patients' bodies do not respond normally to insulin.
- Stress induced from critical illness increase insulin resistance.

Figure: Consequence of insulin resistance in ICU patients.

Insulin Resistance in ICU Patients



- Critically ill patients' bodies do not respond normally to insulin.
- Stress induced from critical illness increase insulin resistance.
- Lack of glycogen production in the liver creates an excess of blood glucose (hyperglycaemia).

Figure: Consequence of insulin resistance in ICU patients.

2026-08-11

Thesis Presentation

Motivation

The Fundamental Problem Addressed

Insulin Resistance in ICU Patients

Insulin Resistance in ICU Patients



- Critically ill patients' bodies do not respond normally to insulin.
- Stress induced from critical illness increase insulin resistance.
- Lack of glycogen production in the liver creates an excess of blood glucose (hyperglycaemia).

Figure: Consequence of insulin resistance in ICU patients.

Addressing Hyperglycaemia

- **Problem:**

- **A solution:**

2026-08-11

Thesis Presentation

└ Motivation

— The Fundamental Problem Addressed

Addressing Hyperglycaemia

Addressing Hyperglycaemia

- **Problem:**
Stress-induced hyperglycaemia **increases** the likelihood of **mortality**.
- **A solution:**

2026-08-11

Thesis Presentation

└ Motivation

└ The Fundamental Problem Addressed

Addressing Hyperglycaemia

- Problem:** Stress-induced hyperglycaemia increases the likelihood of mortality.

Addressing Hyperglycaemia

2026-08-11

Thesis Presentation

- Motivation

└ The Fundamental Problem Addressed

└ Addressing Hyperglycaemia

Addressing Hyperglycaemia

- | | |
|---|---|
| <p>Problem:
Stress-induced hyperglycaemia increases the likelihood of mortality.</p> | <p>A solution:
Maintaining glycaemic levels to 4-6 mmol/L reduced mortalities by 50% (Berghe et al. 2001).</p> |
|---|---|

- **Problem:**
Stress-induced hyperglycaemia **increases** the likelihood of **mortality**.
- **A solution:**
Maintaining glycaemic levels to 4-6 mmol/L reduced mortalities by 50% (Berghe et al. 2001).

Intensive Insulin Therapy

- **What:**
- **Goal:**
- **How:**

2026-08-11

Thesis Presentation

- Motivation

—The Fundamental Problem Addressed

Intensive Insulin Therapy

- What:
- Goal:
- How:

- **What:** Utilizing aggressive insulin infusions and feeding inputs to tightly control blood glucose.
- **Goal:**
- **How:**

- **What:** Utilizing aggressive insulin infusions and feeding inputs to tightly control blood glucose.
- **Goal:** Prevent hyperglycaemia in patients to reduce mortalities and longer visitation stays.
- **How:**

- **What:** Utilizing aggressive insulin infusions and feeding inputs to tightly control blood glucose.
- **Goal:** Prevent hyperglycaemia in patients to reduce mortalities and longer visitation stays.
- **How:** Insulin therapy traditionally has relied on ad-hoc sliding protocols.

2026-08-11

Thesis Presentation

└ Motivation

└ The Fundamental Problem Addressed

└ Intensive Insulin Therapy

- **What:** Utilizing aggressive insulin infusions and feeding inputs to tightly control blood glucose.
- **Goal:** Prevent hyperglycaemia in patients to reduce mortalities and longer visitation stays.
- **How:** Insulin therapy traditionally has relied on ad-hoc sliding protocols.

- **What:** Utilizing aggressive insulin infusions and feeding inputs to tightly control blood glucose.
- **Goal:** Prevent hyperglycaemia in patients to reduce mortalities and longer visitation stays.
- **How:** Insulin therapy traditionally has relied on ad-hoc sliding protocols. (?)

2026-08-11

Thesis Presentation

└ Motivation

└ The Fundamental Problem Addressed

└ Intensive Insulin Therapy

- **What:** Utilizing aggressive insulin infusions and feeding inputs to tightly control blood glucose.
- **Goal:** Prevent hyperglycaemia in patients to reduce mortalities and longer visitation stays.
- **How:** Insulin therapy traditionally has relied on ad-hoc sliding protocols. (?)

Example of an Ad-hoc Sliding Protocol

2026-08-11

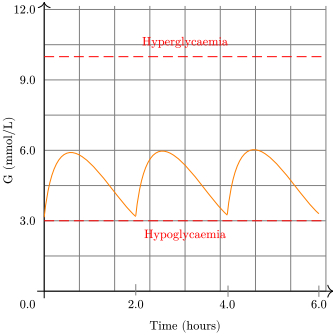
Thesis Presentation

└ Motivation

└ The Fundamental Problem Addressed

└ Example of an Ad-hoc Sliding Protocol

Limitations of Tight Glucose Control



- What:
- Why:

Figure: Tightly Controlled Blood Glucose

2026-08-11

Thesis Presentation

- └ Motivation
- └ The Fundamental Problem Addressed
- └ Limitations of Tight Glucose Control

Limitations of Tight Glucose Control



- What:
- Why:

Figure: Tightly Controlled Blood Glucose

Limitations of Tight Glucose Control

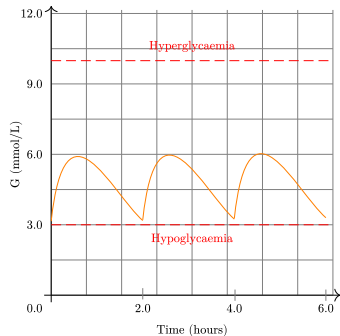


Figure: Tightly Controlled Blood Glucose

- **What:**
Poor patient adaptability results in cases of **hypoglycaemia** and **high glycaemic variation**.
- **Why:**

2026-08-11

Thesis Presentation

└ Motivation

└ The Fundamental Problem Addressed

└ Limitations of Tight Glucose Control



- **What:**
Poor patient adaptability results in cases of **hypoglycaemia** and **high glycaemic variation**.
- **Why:**

Figure: Tightly Controlled Blood Glucose

Limitations of Tight Glucose Control

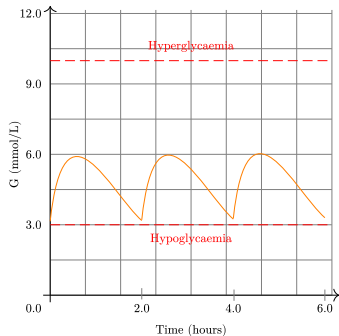


Figure: Tightly Controlled Blood Glucose

- **What:**
Poor patient adaptability results in cases of **hypoglycaemia** and **high glycaemic variation**.
- **Why:**
Solely utilizing insulin infusions struggles to address patient variability.

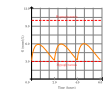
2026-08-11

Thesis Presentation

Motivation

The Fundamental Problem Addressed

Limitations of Tight Glucose Control

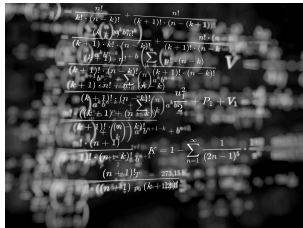


- **What:**
Poor patient adaptability results in cases of **hypoglycaemia** and **high glycaemic variation**.
- **Why:**
Solely utilizing insulin infusions struggles to address patient variability.

Figure: Tightly Controlled Blood Glucose

Infrequent measurements, not considering nutrition, and not complex to monitor dynamics changes in patient health

Role of Model-based Glycaemic Control



Physiological models can provide more insight into patients' insulin-glucose dynamics than solely relying on glycaemic measurements.

2026-08-11

Thesis Presentation

- Motivation

-The Fundamental Problem Addressed

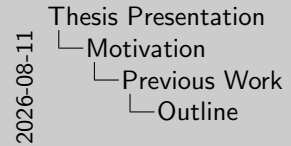
–Role of Model-based Glycaemic Control



Physiological models can provide more insight into patients' insulin-glucose dynamics than solely relying on glycaemic measurements.

Insulin infusions combined with better understanding of dynamics can provide more insight to the clinicians

- 1 Motivation
 - The Fundamental Problem Addressed
 - Previous Work
- 2 Results and Contributions
 - Main Results
 - Key Ideas Behind Implementation
- 3 Summary
 - Takeaways
 - Further Reading



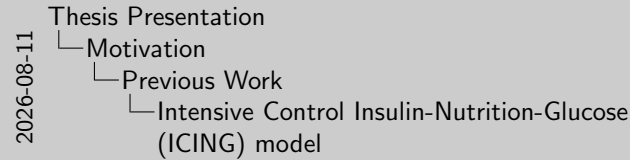
Outline

- Motivation
 - The Fundamental Problem Addressed
 - Previous Work
- Results and Contributions
 - Main Results
 - Key Ideas Behind Implementation
- Summary
 - Takeaways
 - Further Reading

Intensive Control Insulin-Nutrition-Glucose (ICING) model

Lin et al. has developed a clinically-validated nonlinear insulin-glucose model with the following main components

Dynamic	Notation	Value
Blood Glucose	$\dot{G}$	mmol/L
Intersitial Plasma	$\dot{Q}$	mU/L
Plasma Insulin	$\dot{i}$	mU/L
Glucose in the Stomach	$\dot{P1}$	mmol
Glucose in the Gut	$\dot{P2}$	mmol
Endogenous Insulin Production	$u_{en}(t)$	mU/min
Insulin Sensitivity	$S_I(t)$	-
Input	Notation	Value
Exogenous Insulin	$u_{ex}(t)$	mU/min
External Food	$D(t)$	mmol/min
Parenteral Glucose	$P_N(t)$	mmol/min



Intensive Control Insulin-Nutrition-Glucose (ICING) model

Lin et al. has developed a clinically-validated nonlinear insulin-glucose model with the following main components

Dynamic	Notation	Value
Blood Glucose	G	mmol/L
Intersitial Plasma	Q	mU/L
Plasma Insulin	i	mU/L
Glucose in the Stomach	$P1$	mmol
Glucose in the Gut	$P2$	mmol
Endogenous Insulin Production	$u_{en}(t)$	mU/min
Insulin Sensitivity	$S_I(t)$	-
Input	Notation	Value
Exogenous Insulin	$u_{ex}(t)$	mU/min
External Food	$D(t)$	mmol/min
Parenteral Glucose	$P_N(t)$	mmol/min

Computational Approximation of the ICING model

- The ICING model equations are defined in continuous-time
- Discretization of the model is required for the presented estimation methods

For an example dynamic F

$$\begin{aligned}t_k &= t_{k-1} + \Delta \\ \dot{F} &\approx \frac{F(t_k) - F(t_k - \Delta)}{\Delta} \\ F_k &\approx F_{k-1} + \Delta \dot{F}.\end{aligned}$$

2026-08-11

Thesis Presentation

└ Motivation

└ Previous Work

└ Computational Approximation of the ICING model

- The ICING model equations are defined in continuous-time
- Discretization of the model is required for the presented estimation methods

For an example dynamic F

$$\begin{aligned}t_k &= t_{k-1} + \Delta \\ \dot{F} &\approx \frac{F(t_k) - F(t_k - \Delta)}{\Delta} \\ F_k &\approx F_{k-1} + \Delta \dot{F}.\end{aligned}$$

1. According to Euler's method
2. Discretization error is bounded by Δ

The discrete-time dynamic ICING model can be represented in the stochastic state-space.

2026-08-11

- Thesis Presentation
 - Motivation
 - Previous Work
 - Bayesian Filtering

Bayesian Filtering

The discrete-time dynamic ICING model can be represented in the stochastic state-space.

The discrete-time dynamic ICING model can be represented in the stochastic state-space.

State-space Representation

$$x_k \sim p(x_k \mid x_{k-1}, u_{k-1})$$
$$y_k \sim p(y_k \mid x_k)$$

System Variable	Notation
Dynamics	x_k
Inputs	u_k
Measurements	y_k

2026-08-11

Thesis Presentation

- Motivation
 - Previous Work
 - Bayesian Filtering

Bayesian Filtering

The discrete-time dynamic ICING model can be represented in the stochastic state-space.

State-space Representation
$x_k \sim p(x_k \mid x_{k-1}, u_{k-1})$ $y_k \sim p(y_k \mid x_k)$

System Variable	Notation
Dynamics	x_k
Inputs	u_k
Measurements	y_k

1. The measurement often depends on the hospital, but will be discussed more later

The discrete-time dynamic ICING model can be represented in the stochastic state-space.

State-space Representation

$$x_k \sim p(x_k \mid x_{k-1}, u_{k-1})$$
$$y_k \sim p(y_k \mid x_k)$$

System Variable	Notation
Dynamics	x_k
Inputs	u_k
Measurements	y_k

There are various methods of optimal filtering, and each has their own assumptions and limitations.

2026-08-11

Thesis Presentation

- Motivation
 - Previous Work
 - Bayesian Filtering

Bayesian Filtering

The discrete-time dynamic ICING model can be represented in the stochastic state-space.

State-space Representation

$$x_k \sim p(x_k \mid x_{k-1}, u_{k-1})$$
$$y_k \sim p(y_k \mid x_k)$$

System Variable	Notation
Dynamics	x_k
Inputs	u_k
Measurements	y_k

There are various methods of optimal filtering, and each has their own assumptions and limitations.

1. The measurement often depends on the hospital, but will be discussed more later
2. Just the EKF and the PF will be in the topic of this thesis

2026-08-11

- Thesis Presentation
 - Motivation
 - Previous Work
 - Extended Kalman Filter

Outline

1 Motivation

- The Fundamental Problem Addressed
- Previous Work

2 Results and Contributions

- Main Results
- Key Ideas Behind Implementation

3 Summary

- Takeaways
- Further Reading

2026-08-11

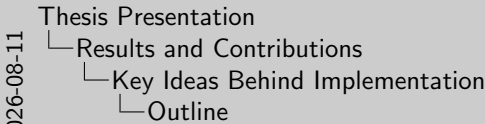
Thesis Presentation

Results and Contributions

- Main Results

–Outline

- 1 Motivation
 - The Fundamental Problem Addressed
 - Previous Work
- 2 Results and Contributions
 - Main Results
 - Key Ideas Behind Implementation
- 3 Summary
 - Takeaways
 - Further Reading



Outline	
Motivation	• The Fundamental Problem Addressed • Previous Work
Results and Contributions	• Main Results • Key Ideas Behind Implementation
Summary	• Takeaways • Further Reading

Outline

1 Motivation

- The Fundamental Problem Addressed
- Previous Work

2 Results and Contributions

- Main Results
- Key Ideas Behind Implementation

3 Summary

- Takeaways
- Further Reading

2026-08-11

- Thesis Presentation
 - Summary
 - Takeaways
 - Outline

Outline

- Motivation
 - The Fundamental Problem Addressed
 - Previous Work
- Results and Contributions
 - Main Results
 - Key Ideas Behind Implementation
- Summary
 - Takeaways
 - Further Reading

Outline

1 Motivation

- The Fundamental Problem Addressed
- Previous Work

2 Results and Contributions

- Main Results
- Key Ideas Behind Implementation

3 Summary

- Takeaways
- Further Reading

2026-08-11

- Thesis Presentation
 - Summary
 - Further Reading
 - Outline

 Berghe, Greet Van den et al. (Nov. 2001). “Intensive Insulin Therapy in Critically Ill Patients”. In: DOI: 10.1056/NEJMoa011300.

 Lin, Jessica et al. (May 2011). “A physiological Intensive Control Insulin-Nutrition-Glucose (ICING) model validated in critically ill patients”. In: DOI: 10.1016/j.cmpb.2010.12.008.

2026-08-11

- Thesis Presentation
 - Summary
 - Further Reading
 - References

References

-  Berghe, Greet Van den et al. (Nov. 2001). “Intensive Insulin Therapy in Critically Ill Patients”. In: DOI: 10.1056/NEJMoa011300
-  Lin, Jessica et al. (May 2011). “A physiological Intensive Control Insulin-Nutrition-Glucose (ICING) model validated in critically ill patients”. In: DOI: 10.1016/j.cmpb.2010.12.008.